

Lecture 13: Shrinkage Estimation

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13.1 A Prediction Contest

Suppose an early leaderboard reports noisy estimates for 30 players from limited samples. For example, each estimate could be a batting average after a small number of at-bats or a shooting percentage after a small number of attempts. The task is to submit 30 predictions of the players' underlying abilities. The score is total squared error:

$$L(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}) = \sum_{i=1}^k \left(\theta_i - \hat{\theta}_i \right)^2.$$

Which predictions minimize expected contest loss?

13.2 The Observed Leaderboard

Let X_i be player i 's observed early estimate and θ_i be the player's underlying ability on the same scale. Assume

$$X_i \sim \mathcal{N}(\theta_i, \sigma^2), \quad i = 1, \dots, k,$$

independently, where the unknown θ_i are fixed constants. In vector notation,

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 I).$$

The usual estimator is the MLE:

$$\hat{\boldsymbol{\theta}}^{\text{MLE}} = \mathbf{X}.$$

It is unbiased coordinate by coordinate, but it treats every observed extreme as entirely real. Players at the top of a noisy leaderboard tend to combine high ability with favorable noise; players at the bottom tend to combine low ability with unfavorable noise.

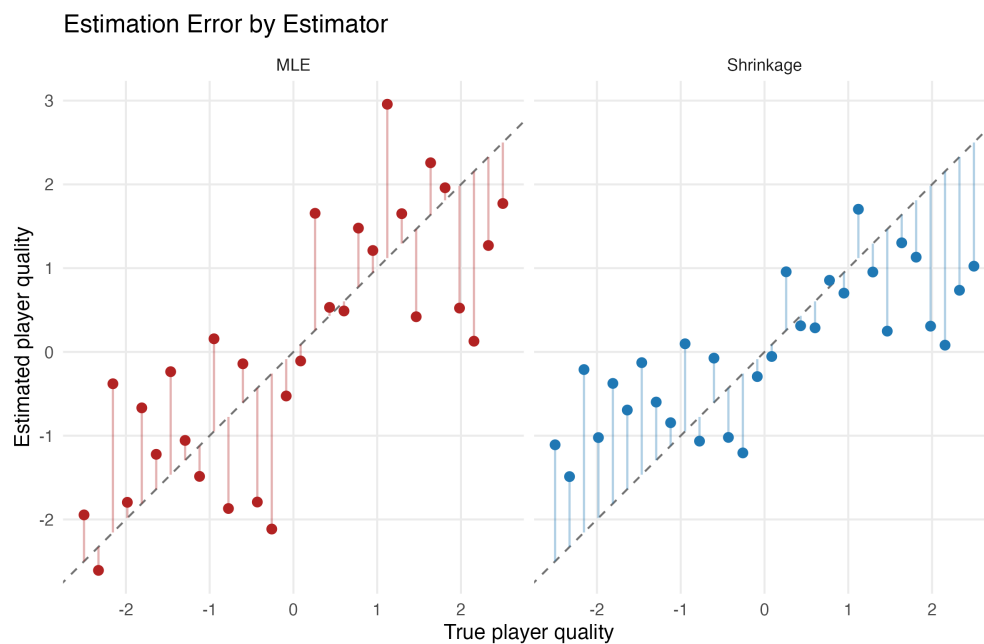


Figure 13.1: True and estimated player quality under the MLE and shrinkage estimator.

13.3 A Shrunk Leaderboard

A simple alternative keeps only a fraction b of each player's observed distance from the mean:

$$\hat{\theta}_i(b) = \bar{X} + b(X_i - \bar{X}), \quad 0 \leq b \leq 1.$$

The endpoints recover two familiar strategies:

- $b = 1$: no pooling, so $\hat{\theta}_i = X_i$;
- $b = 0$: complete pooling, so every estimate equals $\bar{X}$;
- $0 < b < 1$: partial pooling toward $\bar{X}$.

A value such as $b = 0.7$ retains 70% of every observed difference from the mean. The remaining question is how to choose b .

13.4 Risk: The Contest Score

Definition 13.1 (Risk). For an estimator $\hat{\theta}$ and a loss function $L(\theta, \hat{\theta})$, the risk is expected loss:

$$R(\theta, \hat{\theta}) = \mathbb{E}_{\theta} [L(\theta, \hat{\theta})].$$

In this contest, L is total squared error. For the MLE,

$$R(\theta, \mathbf{X}) = k\sigma^2.$$

Every coordinate contributes variance σ^2 . A shrinkage estimator introduces some bias but can reduce enough variance to lower the total risk under this loss function. The simulation below reports mean squared error, which is total squared error divided by the fixed number of players. This rescaling does not change which estimator has lower loss.

13.5 Data-Adaptive Shrinkage

A positive-part James–Stein-style estimator chooses the multiplier from the observed spread of the leaderboard:

$$\hat{\theta}_i^{\text{shrink}} = \bar{X} + \left[1 - \frac{(k-3)\sigma^2}{\sum_{j=1}^k (X_j - \bar{X})^2} \right]_+ (X_i - \bar{X}),$$

where $[a]_+ = \max(a, 0)$.

The multiplier determines the amount of shrinkage:

- a spread-out leaderboard is treated as evidence of real differences, so shrink less;
- a tightly clustered or very noisy leaderboard is shrunk more strongly.

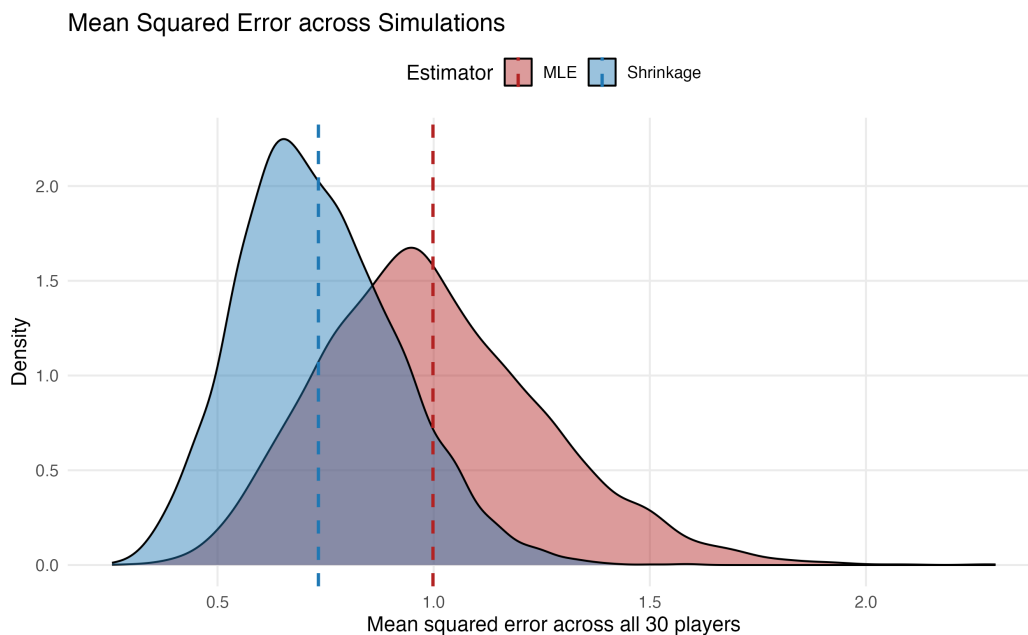


Figure 13.2: Mean squared error across 5,000 simulated leaderboards.

13.6 James–Stein Dominance

Theorem 13.2 (James–Stein and Centered Shrinkage, informal). *When estimating $k \geq 3$ independent normal means with common known variance under total squared-error loss, classic James–Stein shrinkage*

toward a fixed target can dominate $\mathbf{X}$. When the target is the estimated grand mean $\bar{X}$, one dimension is used to estimate that center; the corresponding centered dominance result requires $k \geq 4$.

The theorem does *not* say:

- every shrinkage rule beats the MLE;
- shrinkage is better for every player in every realized dataset;
- the group mean is always the right target;
- squared error is always the right loss function.

It says that an apparently natural unbiased estimator can be improved when the decision is evaluated jointly.

13.7 Individual and Joint Performance

Suppose a general manager asks for one player's estimate, while the analytics department is evaluated on predictions for the entire league. Those are different decision problems.

Example 13.3 (A Real Breakout and a Noisy Leaderboard). Suppose Player 7 is a real breakout. Shrinkage pulls the estimate downward and makes that player's prediction worse. Player 13 appears extreme mostly because of noise, so shrinkage improves that prediction substantially. The department's total squared error can fall even though its advice about Player 7 gets worse.

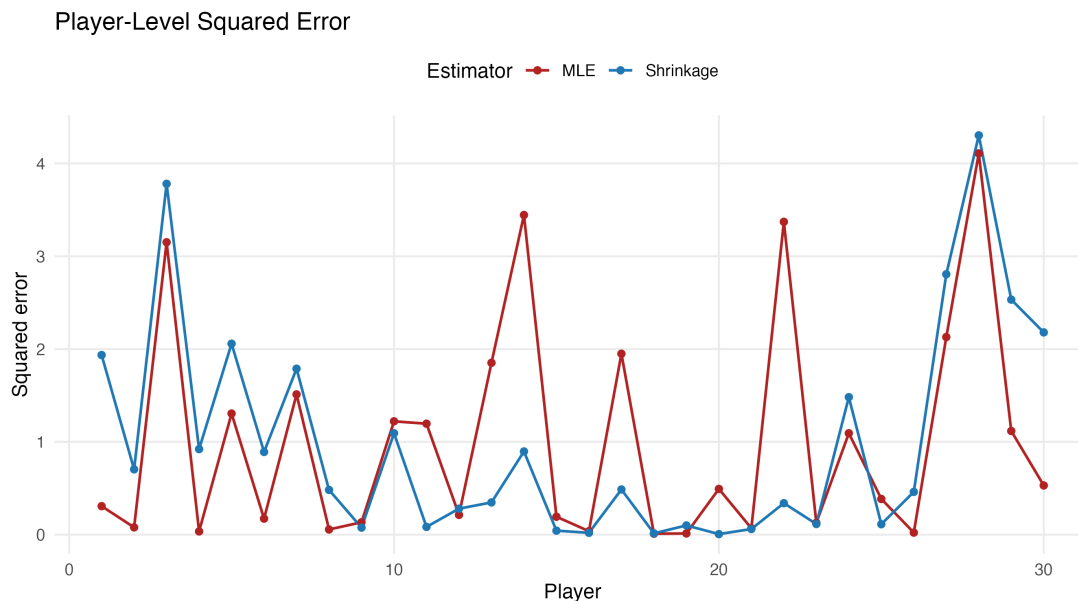


Figure 13.3: Player-level squared error for the MLE and shrinkage estimator.

A joint-risk guarantee does not establish a coordinate-by-coordinate guarantee. For one player named before seeing the data, James–Stein does not say shrinkage is better. Shrinkage toward m helps that player when

the true value is close enough to m , and hurts when the player is genuinely far from m . If the player was selected because of an extreme observed estimate, the case for shrinkage is stronger because selection makes noise part of the story.

The correct estimator also depends on which errors matter, how errors are weighted, and whether overestimation and underestimation have different costs.

The loss function is not bookkeeping added after estimation. It defines what “better” means for the decision being scored.

13.8 Why Shrinkage Reduces Risk

Observed extremes are selected partly for favorable or unfavorable noise. The sampling model describes $\mathbb{E}[X_i | \theta_i] = \theta_i$, while prediction asks what values of θ_i are plausible after observing X_i . These regression directions are not generally the same.

The MLE avoids bias but can have high variance. Shrinkage accepts controlled bias to reduce variance, and total squared-error loss scores the full set of estimates as one decision. Information from the entire leaderboard can therefore improve the combined action.

When observations have different known standard errors, the same framework can be used approximately after standardization:

$$Z_i = \frac{X_i - m}{\sigma_i}, \quad \eta_i = \frac{\theta_i - m}{\sigma_i}, \quad Z_i \sim \mathcal{N}(\eta_i, 1).$$

Here m is a meaningful common target, such as the league mean.

13.9 Shrinkage Across Statistical Methods

The same shrinkage shape appears in several frameworks:

- **Bayes:** a prior pulls estimates toward its center.
- **Empirical Bayes:** the population estimates the center and strength.
- **James–Stein:** a frequentist risk calculation justifies shrinkage under joint loss.
- **Ridge regression:** a penalty shrinks coefficients toward zero.

These methods are not identical. Each reduces the variability of noisy estimates by accepting some bias.

13.10 Takeaways

- Shrinkage can be justified by frequentist risk, not only by a prior.
- James–Stein concerns joint estimation under a specific loss function.
- Better aggregate performance does not mean better performance for every individual.
- The shrinkage target, strength, and loss function must match the decision.

Appendix

A Suggested Reading

- James and Stein, “Estimation with Quadratic Loss,” for the original inadmissibility result.
- Stephen Stigler, “A Galtonian Perspective on Shrinkage Estimators,” for the regression interpretation.